6 Trigonometry

Exercise 6.1

1. $\sin B = \frac{9}{41}; \cos B = \frac{40}{41}; \tan B = \frac{9}{40};\ \text{cosec}B = \frac{41}{9}; \sec B = \frac{41}{40}; \cot B = \frac{40}{9}$

2. (i) $\sin B = \frac{12}{13}$ (ii) $\sec B = \frac{13}{5}$ (iii) $\cot B = \frac{5}{12}$ (iv) $\cos C = \frac{4}{5}$
(v) $\tan C = \frac{3}{4}$ (vi) $\text{cosec}C = \frac{5}{3}$

3. $\sin\theta = \frac{1}{2}; \cos\theta = \frac{\sqrt{3}}{2}; \tan\theta = \frac{1}{\sqrt{3}};\ \text{cosec}\theta = \frac{2}{1}; \sec\theta = \frac{2}{\sqrt{3}}; \cot\theta = \sqrt{3}$

4. $\frac{3}{40}$ 5. $\sin A = \frac{1-x^2}{1+x^2}; \tan A = \frac{1-x^2}{2x}$ 7. $\frac{1}{2}$

8. $\frac{1}{2}$ 9. $\sin\alpha = \frac{4}{5}; \cos\beta = \frac{4}{5}; \tan\phi = \frac{4}{3}$ 10. 7m

Exercise 6.2

2.(i) 0 (ii) $\frac{7}{4}$ (iii) 3 4. 2

Exercise 6.3

1.(i) 1 (ii) 1 (iii) 1 (iv) 2

Exercise 6.4

1.(i) 0.7547 (ii) 0.2648 (iii) 1.3985 (iv) 0.3641
(v) 0.8302 (vi) 2.7907 2.(i) 85°57′ (or) 85°58′ (or) 85°59′
(ii) 47°27′ (iii) 4°7′ (iv) 87°39′ (v) 82°30′
3.(i) 1.9970 (ii) 2.8659 4. 18.81 cm^2 5. 36°52′
6. 54.02 m

Exercise 6.5

1. (1) 2. (2) 3. (2) 4. (3) 5. (2) 6. (3) 7. (3) 8. (1) 9. (2) 10. (2)

7 Mensuration

Exercise 7.1

1.(i) 120 cm^2 (ii) 7.2 m^2 2. 1320 m^2, ₹26400 3. 12000 m^2
4. 1558.8 cm^2 5. ₹ 1050 6. 240 cm^2 7. 138 cm^2
8. 354m^2 9. 1536 m^2 10. 672 m^2

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